Messing About With Figures

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At Horizon, our investment decisions are rarely predicated on purely macroeconomic views; however, like John Maynard Keynes, we tend to derive a variety of interesting macroeconomic insights from simply "messing about with figures and seeing what they must mean."[1] While this exercise in explicating the logical relations among certain variables is merely one element of a rigorous investment process, it highlights the importance of analyzing primary source data as opposed to the alarmist and frequently misleading content one might encounter in the media.

The following excerpts from recent Horizon research reports explore two issues oft mentioned by the financial media: the American Recovery and Reinvestment Act of 2009 and home prices. The difference, however, is that they do so from a primary source perspective and lead one to seriously, and necessarily, question the conventional wisdom.[2]

Facts & Figures - The American Recovery and Reinvestment Act of 2009[3]

The following facts and figures come from Recovery.gov, the official Federal Government website dedicated to providing statistics on the monies that have been spent under the American Recovery and Reinvestment Act of 2009 (the "Act").[4] As part of the $787 billion appropriated under the Act, $275 billion was appropriated for distribution by federal departments and agencies through contracts, grants, and loans. This is, so to speak, the classical Keynesian portion of the bill. An important part of the Act is Section 1512, also known as the Jobs Accountability Act, which governs the disclosure of funds awarded and received.

It is important to note the difference between the terms "awarded" and "received": "awarded" means that the funds have been earmarked, in other words, by one of 28 government agencies responsible for disbursing the funds appropriated by Congress; only when the funds have been designated as "received" do the ultimate recipients actually have the funds at their disposal. Furthermore, it is important to remember that in many cases these are construction projects, and the funds are expended in accordance with a draw schedule as the funds are needed. So, to a certain degree it is up to the recipients as to when "received" funds are actually spent on creating additional economic activity.

According to Recovery.gov, as of September 30, 2010, the funds awarded through contracts, grants, and loans amounted to approximately $251 billion, or about 32 percent of the $787 billion stimulus package and 91 percent of the $275 billion dollar appropriation.[5] More importantly, of the $251 billion in awarded funds, only $112 billion have actually been received by the states—that is, only 40 percent of the $275 billion appropriated to contracts, grants, and

[1] Robert Skidelsky, *John Maynard Keynes: The Economist as Savior 1920-1937*, pg. xix.

[2] The content herein was originally presented as a series of speeches made by Murray Stahl during the third quarter of 2010. The data presented in the original presentations has been updated.

[3] *The Fixed Income Contrarian Compendium*, *Facts & Figures* section, September 2010.

[4] The American Recovery and Reinvestment Act of 2009:
http://frwebgate.access.gpo.gov/cgi-bin/getdoc.cgi?dbname=111_cong_bills&docid=f:h1enr.pdf

[5] Award recipients report every quarter on their awards. Reporting for Q3 2010 ended on October 29, 2010. Recipients will continue to correct reports if necessary until December 6, 2010. Final data will be published on December 8, 2010. Source: http://www.recovery.gov/Pages/TextView.aspx?data=recipientFundingAwardedByState

loans have actually been distributed to their end users, representing just over 44 percent of total awarded funds.

The table below shows the funds that have been awarded to several of the larger states, as well as those that were received from February 17, 2009 through September 30, 2010. On average, for the top 10 states by 2008 gross state product ("GSP"), approximately 44 percent of awarded funds have actually been received. Interestingly, for the top 10 states listed below, awarded funds represent approximately 1.5 percent of GSP, in most cases with very little variation.

USD Millions	Funds Awarded	Funds Received	Funds Received as % of Funds Awarded	Nominal Gross State Product (2008)	Funds Awarded as % of GSP
California	$27,132	$12,835	47.3%	1,846,757	1.5%
Texas	$16,392	$6,658	40.6%	1,223,511	1.3%
New York	$15,467	$6,056	39.2%	1,144,481	1.4%
Florida	$10,927	$4,269	39.1%	744,120	1.5%
Illinois	$10,218	$5,271	51.6%	633,697	1.6%
Pennsylvania	$9,185	$4,143	45.1%	553,301	1.7%
New Jersey	$5,370	$2,868	53.4%	474,936	1.1%
Ohio	$8,594	$3,307	38.5%	471,508	1.8%
North Carolina	$7,303	$2,771	37.9%	400,192	1.8%
Georgia	$6,640	$3,361	50.6%	397,756	1.7%
Average, Top 10	n/a	n/a	44.3%	n/a	1.5%

Source: Recovery.gov, U.S. Department of Commerce

California, for example, has been awarded $27.1 billion, but has received only $12.8 billion. Again, merely because the money is received doesn't mean that it has actually been spent. Therefore, from the point of view of the data presented on Recovery.gov as mandated by Section 1512 of the Act, a very small portion of the funds have actually been spent.

Legally, according to the provisions of the Act, the funds must be spent no later than the end of February 2012. It must logically follow that if the government is going to abide by the provisions of its own enactment, a great quantity of money will be spent between now and February 2012 (i.e., approximately 59 percent of $275 billion, or $163 billion).[6]

Murray's Musings - Has the Housing Market Already Experienced Deflation?[7]

This section pertains not only to the most recent residential sales figures that were released on October 27, 2010, but also to some historical figures. According to the U.S. Census Bureau, the seasonally adjusted annual rate of new home sales in September 2010 was 307,000 homes per annum. That can be compared to the revised August 2010 seasonally adjusted annualized number of 288,000 homes per annum. These figures are about the lowest on record. With respect to prices, in September 2010, the median sales price for a new home was $223,800, whereas the average sales price was $257,500.[8]

[6] These figures only address the $275 billion appropriated under the contracts, grants, and loans portion of the Act.
[7] *The Stahl Report Compendium*, *Murray's Musings* section, August 2010.
[8] Source: http://www.census.gov/const/www/newressaleshist.html

Putting this in an historical context, the seasonally adjusted annual rate of new home sales in September 2002—chosen because this is during the recession caused by the collapse of the internet bubble but prior to what became the housing bubble—was 1,054,000 homes per annum. The median sales price of a home was $176,300, whereas the average sales price was $218,100.

New Residential Sales

	New Single-Family Homes*	Median Price	Average Price
September 2010	307,000	$223,800	$257,500
August 2010	288,000	$204,700	$248,800
September 2002	1,054,000	$176,300	$218,100
September 1996	782,000	$137,500	$171,600

*seasonally adjusted annual rate
Source: U.S. Census Bureau

Comparing the September 2002 sales price to the September 2010 sales price, the compound annual rate of change on an average price basis shows an increase of 2.1 percent per year, while the median price shows an increase of 3.0 percent per year. If the September 2002 data represents normal prices for a recession and if one takes into account the annualized rate of inflation of 2.4 percent over the same period, then one might conclude that we've effectively returned to pre-bubble prices.

Going back even further to September 1996, which is clearly a period that precedes both the housing bubble and the internet bubble, the seasonally adjusted annual rate of new home sales was 782,000 homes per annum. As well, one must remember that the population in the United States was 15 percent lower at that point. At the same time, the median sales price of a home was $137,500, whereas the average sales price was $171,600.

Comparing these historical figures to current prices results in an annualized increase in home prices of 2.9 percent based on the average price. If, however, one takes into account the rate of inflation of 2.3 percent per annum over the same period, since 1996, the average price of a home has increased only marginally in real terms. Such comparisons merely on the basis of price changes are not ideal, however, and can prove to be misleading. Since mortgage financing is a central factor in the purchase of most homes, one must take into account mortgage rates as well as home prices.

According to the Federal Reserve's website, in September 1996, the conventional 30-year mortgage rate was 8.23 percent. According to the Mortgage Bankers Association, as of November 5, 2010, the conventional 30-year mortgage rate was 4.28 percent, and the rate for a 15-year mortgage was 3.64 percent.

Let us assume that in September 2010 one wished to borrow money in the standard way and buy an average-priced home selling for $257,500. If one borrowed 80 percent of the selling price, which is $206,000, at the 30-year rate of 4.28 percent, the monthly payment would be $1,017.02. Alternatively, if one bought an average-priced home in September 2010, but borrowed at the 30-year rate that prevailed in September 1996 (i.e., 8.23 percent), the monthly payment would be

$1,544.71, over 50 percent more than at the lower rate. That difference gives you an idea of the degree to which lower interest rates affect the cost of buying a new home.

Taking the $171,600 average price of a home in September 1996, borrowing 80 percent of that sum, which is $137,280, and financing the balance at the then conventional 30-year mortgage rate of 8.23 percent would result in a monthly payment of $1,029.41. This is virtually identical to the current monthly payment on a more expensive home financed at a lower rate (i.e., $257,500 at 4.28 percent).

One might reasonably conclude that housing is now effectively available at the September 1996 levels, factoring in mortgage rates, but not the cost of insurance and property taxes. Therefore, one shouldn't anticipate deflation in housing, because it has already been achieved. One can actually buy a home for a monthly payment that is not very many dollars different from the monthly payment one would have needed in September 1996 when rates were significantly higher.

In other words, if you took the $1,017.02 monthly payment as it existed in September 2010 and adjusted it for inflation over the same 14 year period, the real payment would be $734.69 (i.e., 1996 prices). So, in real terms, the monthly payment on the average-priced home is nearly 30 percent lower today than it was 14 years ago.

Final Thoughts

The consultation of secondary sources might lead one to conclude that the American Recovery and Reinvestment Act has failed as a tool of fiscal policy and that home prices are on an inexorable march towards disaster. The original source data, however, indicates that a disproportionate amount of funds awarded under the Act have yet to be spent. Therefore, such conclusions appear rather premature.

Moreover, while it is possible that bank liquidations of portfolios of foreclosed homes could have a negative short-term impact on home prices, current home prices and mortgage rates suggest that much of the deflationary risk has already been taken into account. In other words, the long-term risk of a continued diminution in aggregate home prices appears lower than one might conclude after consulting secondary sources.